

Experiment 7: Logical and Probabilistic Reasoning

Aim

To implement basic Propositional Logic, First-Order Logic, and probability-based reasoning using Python.

Exercises

1. Propositional Logic

Represent a simple rule such as “If it rains, then the road is wet” using Propositional Logic. Generate its truth table using Python and verify the result.

2. First-Order Logic

Represent the statements “All humans are mortal” and “Socrates is a human” using First-Order Logic. Write a simple Python program to derive whether Socrates is mortal.

3. Probability of Events

Consider a small dataset of events such as students who passed/failed and attended/did not attend. Calculate the probability of individual events, joint probability, and probability of the union of two events using Python.

4. Conditional Probability

Using the same dataset, calculate conditional probabilities such as $P(A | B)$ and $P(B | A)$. Display the calculated probabilities using Python.

5. Bayes' Theorem

Implement a simple Bayes' theorem example, such as disease diagnosis. Calculate the probability of a disease after observing a positive test result.

6. Visualization

Use Matplotlib to create a suitable graph showing the calculated probabilities. Modify one of the input probabilities and observe how the graph changes.